

Enhancing Spatial and Temporal Coverage of Soil Moisture Estimation Using Satellite and Weather-Driven Machine Learning

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Abstract—Soil moisture monitoring is essential for hydrology, agriculture, and climate science, but current methods are limited in either spatial or temporal coverage. In-situ sensors provide accurate, high-frequency measurements but are sparsely deployed, while satellite observations offer broader spatial coverage but often lack daily continuity. To address this gap, we propose a satellite-driven machine learning approach that uses data imputation and XGBoost to estimate daily soil moisture at locations without sensors. Model inputs include Sentinel-1, Sentinel-2, Moderate Resolution Imaging Spectroradiometer (MODIS), Soil Moisture Active Passive (SMAP), a digital elevation model, and weather variables, with ground-truth soil moisture from five public stations in Washington State, USA. Temporal evaluation achieved strong performance, with $R^2 = 0.822$, $MAE = 0.0283$, $RMSE = 0.0397$, $ubRMSE = 0.0396$, $bias = -0.0032$, and the 90th quantile of residuals = 0.0615. Spatial evaluation showed moderate performance with slightly higher errors due to station-to-station variability. This work is part of an ongoing interdisciplinary effort to develop low-cost, ground-based soil moisture sensors for resource-constrained environments and to design data fusion approaches that enhance the spatial and temporal coverage of soil moisture estimation. Future work will focus on integrating these new in-situ measurements to further evaluate the data fusion framework.

Keywords—soil moisture, remote sensing, satellite observations, machine learning, in-situ sensing

I. INTRODUCTION

Soil moisture is a key factor in hydrology, agriculture, and climate science, as it governs water and energy exchanges at the land-atmosphere interface and strongly affects vegetation growth and the occurrence of droughts. Accurate and high-resolution soil moisture data are essential for applications such as irrigation management, flood and drought monitoring and

climate prediction. Moreover, soil moisture variability affects water resource planning and ecosystem services, making continuous estimation essential across space and time [1]. In resource-constrained environments, the effects of limited and unreliable soil moisture information are often disproportionately severe, because communities may already face limited access to irrigation infrastructure, fewer monitoring systems, and reduced capacity to respond effectively to drought conditions [2] [3].

Despite advances in measurement technologies, obtaining soil moisture at both high spatial and temporal resolution remains challenging. Typical in-situ soil moisture measurements include direct methods such as gravimetric sampling and electromagnetic sensors, including Time Domain Reflectometry (TDR), Frequency Domain Reflectometry (FDR), and capacitance probes [4]. These methods provide high temporal resolution and accurate point measurements but are costly to deploy and maintain, resulting in sparse spatial coverage. On the other hand, satellite remote sensing missions, including Soil Moisture Active Passive (SMAP), Soil Moisture and Ocean Salinity mission (SMOS), Sentinel 1 and Sentinel 2, offer broad spatial coverage and multiple observations per week, but they are not available daily (e.g. SMAP has ~ 3-day global revisit cycle, Sentinel-1 has ~ 6 – 12 days revisit cycle), limiting direct daily soil moisture estimation [5].

To overcome these limitations, researchers have increasingly turned to using remote satellite data to train machine learning models for soil moisture estimation. Many machine learning studies aim to merge multi-source satellite data, but most implicitly rely on satellite revisit schedules (e.g. ~ 3 days for SMAP) and do not explicitly fill temporal gaps and systematically address finding daily soil moisture values when satellite observations are missing [6] [7] [8]. SoMo.ml [9] trains a Long Short-Term Memory (LSTM) model on in-site

measurements from $> 1,000$ stations to estimate daily soil moisture data. It performs well temporally via cross-validation and intercomparison with existing soil moisture datasets. However, it does not evaluate the model's spatial generalizability. On the other hand, some spatial downscaling studies evaluate general spatial performance [6], but they rarely perform tests using custom sensor deployment in completely unseen locations. Few studies provide combined metrics for both temporal forecasting accuracy and spatial generalization tests at new locations for daily soil moisture estimation. Most research uses public in-site networks such as International Soil Moisture Network (ISMN) or large curated station sets, which provide good training and validation for general modeling but do not include plans for custom sensor deployment to test models in truly new, unseen environments.

Moreover, most existing approaches utilize satellite data for soil moisture estimation [10] [11], but weather drivers like precipitation and rain are rarely the inputs for prediction models. Satellite-only inputs without weather variables do not capture the driving meteorological processes that govern soil moisture dynamics across time. Some very recent work [12] explores combining remote sensing data with weather inputs for soil moisture predictions and evaluates spatiotemporal generalization but lacks explicit gap imputation and does not perform tests using custom sensor deployment.

Although existing studies demonstrate the value of these fusion methods for enhancing soil moisture estimation accuracy and resolution, relatively few explicitly address the design and deployment challenges of soil moisture monitoring in resource-constrained environments, where affordability, limited infrastructure, and unreliable power or connectivity can strongly shape what is feasible in practice [13].

To address these challenges, we present a satellite- and weather-driven machine learning approach that integrates multi-source remote sensing and meteorological data to estimate daily soil moisture at locations without in-situ sensors, thereby enhancing both spatial and temporal coverage. The multi-source satellite data includes Sentinel-1, Sentinel-2, Moderate Resolution Imaging Spectroradiometer (MODIS), SMAP, and a digital elevation model (DEM). The weather data includes rain and total precipitation. A data imputation strategy was developed to fill gaps when satellite data are not available, enabling continuous daily soil moisture estimation. We trained an XGBoost [14] model using satellite data and weather data as inputs and using soil moisture measurements from publicly available stations in Washington State (WA), USA as ground truth. To address the broader challenge of deploying in-situ soil moisture sensors in resource-constrained environments, we present the design of a low-cost long-range sensor node that communicates over the public Helium network.

This study is part of an ongoing interdisciplinary collaboration, in which custom soil moisture sensors are being designed and deployed in Seattle, WA. While the current work focuses on model development and evaluation using public sensor data, future integration of newly deployed sensors will enable extensive testing at completely new and unseen locations, extending the model's utility for operational environmental monitoring.

This work has the following contributions:

1. Combining multi-source satellite and weather data to enable soil moisture estimation at locations that do not have sensors deployed, improving spatial coverage.
2. Implementing a data imputation strategy to provide daily soil moisture estimates, enhancing temporal coverage.
3. Evaluating model performance along both temporal and spatial dimensions, demonstrating strong temporal predictive performance and moderate spatial generalization.
4. Designing a low-cost long-range sensor node that will provide in-situ soil moisture measurements for real-world deployment in resource-constrained environments.

The remainder of this paper is organized as follows. Section II presents the data fusion approach, including the datasets used in this study, the data imputation strategy, feature construction, feature engineering and selection, model training using XGBoost, and model evaluation. Section III describes the LoRa-based sensor node, including its design, communication network, experimental setup, sensor calibration, and data collection. Section IV presents the results, interprets the model's temporal and spatial performance as well as feature importance, and discusses in-situ sensor observations. Finally, section V concludes the paper and outlines directions for future work.

II. DATA FUSION APPROACH

A. Dataset

1) Study Area and In Situ Soil Moisture Dataset

The study area is in WA, USA. The ground-truth soil moisture measurements for training and validating models were collected from public in-situ monitoring stations. Five stations in WA were used, including Darrington-21-NNE, Quinault-4-NE, and Spokane-17-SSW from U.S. Climate Reference Network (USCRN) [15], as well as SourdoughGulch_WA_985 and Touchet_WA_824 from a mountain-based monitoring network Snow Telemetry (SNOTEL) [16]. These five stations span the state's major climate regimes, from the wet maritime conditions of the western Cascades and Olympic Peninsula to the semi-arid continental interior of eastern Washington. The temporal coverage of the soil moisture data available from stations is from 2008 to 2025, but some satellite or weather data are not always available in the period. To ensure consistent availability of satellite and weather data, particularly SMAP-derived data, the final study period spanned from January 2017 to December 2025. Given the penetration ability of Sentinel-1 (C-band) radar, the 5 cm soil moisture data were used as ground truth in this study.

2) Satellite and Weather Data

The data used for training the machine learning model includes weather data, satellite data, and terrain information. The raw covariates before feature engineering and feature selection are summarized in Table I.

Weather variables were obtained from the Open-Meteo historical archive API [17]. The raw weather variables used were rain and total precipitation (in millimeters). This data

source was selected due to its availability of hourly observations without requiring extensive interpolation. Hourly values were aggregated into daily totals and were used in training.

Satellite-derived data were collected using Google Earth Engine [18]. The data sources include Sentinel-1, Sentinel-2, MODIS, SMAP, and DEM. Sentinel-1 synthetic aperture radar (SAR) [19] data were used to extract VV and VH backscatter in both decibel and linear representations. Sentinel-2 Level-2A [20] surface reflectance data provided spectral bands B2, B3, B4, B8, B11, and B12. MODIS products [21] were used to obtain land surface temperature (LST) and Normalized Difference Vegetation Index (NDVI). SMAP [22] provided AM and PM soil moisture estimates along with retrieval quality flags. Terrain-related variables were derived from the Shuttle Radar Topography Mission (SRTM) DEM [23], including elevation, slope, and aspect. SoilGrids 2.0 [24] was used to collect soil type.

B. Method

1) Data preprocessing

The data preprocessing pipeline transformed multi-source data into a temporally and spatially aligned dataset for model training by integrating satellite-derived features, weather variables, and in-situ measurements. Besides handling differences in temporal resolution and spatial resolution, it also involved data imputation to fill missing values in satellite data

and applied smoothing to reduce noise in time series signals.

a) Temporal Alignment and Spatial Alignment

Satellite observations, weather data, and in-situ soil moisture measurements are collected at different times and frequencies. To ensure consistency across data sources, all variables were harmonized to a common daily time scale so that satellite- and weather-derived features could be paired with ground-truth soil moisture observations. Ground-truth soil moisture measurements were initially processed at their native temporal resolution. For USCRN stations, soil moisture is provided at standard hourly intervals. SNOTEL stations contain timestamped sub-daily sensor observations. When multiple sensors were available at the same depth, duplicate measurements at identical timestamps were averaged to generate a single station-level value for each time step. USCRN hourly soil moisture values are derived from underlying 5-minute measurements, whereas SNOTEL soil moisture values are reported as instantaneous observations within the Natural Resources Conservation Service system [15] [16]. To ensure consistency across station types, all within-day soil moisture observations were aggregated using the daily mean. This approach produces a unified daily ground-truth soil moisture series for modeling, reduces short-term measurement noise, and aligns the target variable with the daily resolution of the input features.

Weather variables were obtained at an hourly resolution. To ensure temporal alignment with in-situ soil moisture, rainfall and total precipitation were separately aggregated into daily totals (mm). Satellite observations are often sparse and irregular in time. For example, Sentinel-1 SAR does not provide valid measurements for every day at a given station, because revisit timing and acquisition availability depend on orbital coverage and filtering conditions. To match these data with the daily ground-truth soil moisture series, satellite observations within a short temporal window around each day were collected and combined to produce a single representative value for that day. Due to irregular satellite revisit times, multiple observations may fall within this window; when this occurs, their values are combined using a simple average to obtain a daily estimate. The same alignment strategy was applied across all satellite sources in this study. Besides temporal alignment, we also performed spatial alignment to improve robustness to noise and reduce spatial misalignment among datasets. Different satellite datasets have different spatial resolution and footprints – Sentinel-1 SAR has ~ 10 m, Sentinel-2 MSI has 10 – 20 m, MODIS has 250 – 1000 m, SMAP has 9 km, and SRTM DEM has 30 m, and an in-situ station only has a fixed coordinate point. To construct satellite-derived features, a 1km radius buffer was defined around each station. All pixels whose centers fall within this buffer were selected. Then the satellite-derived data (e.g. VV, VH) were averaged to produce a single representative feature for that location.

b) Data Imputation

The datasets used in this study have differing temporal resolutions, which can lead to missing values when constructing a daily dataset. Satellite observations vary by product: Sentinel-1 SAR provides ~6–12 day revisit times depending on location,

TABLE I RAW COVARIATES RETRIEVED FROM SATELLITE AND WEATHER

Raw Covariates	Data Source
Rain	Open-Meteo historical archive API [17]
Total precipitation	Open-Meteo historical archive API [17]
Backscattering coefficients of VV band (VV)	Sentinel-1 synthetic aperture radar (SAR) [19]
Backscattering coefficients of VH band (VH)	Sentinel-1 synthetic aperture radar (SAR) [19]
Blue band of Sentinel-2 (B2)	Sentinel-2 Level-2A [20]
Green band of Sentinel-2 (B3)	Sentinel-2 Level-2A [20]
Red band of Sentinel-2 (B4)	Sentinel-2 Level-2A [20]
Near-infrared band of Sentinel-2 (B8)	Sentinel-2 Level-2A [20]
Shortwave-infrared 1 band of Sentinel-2 (B11)	Sentinel-2 Level-2A [20]
Shortwave-infrared 2 band of Sentinel-2 (B12)	Sentinel-2 Level-2A [20]
LST	MODIS [21]
NDVI	MODIS [21]
AM and PM soil moisture estimates	SMAP [22]
Elevation	SRTM DEM [23]
Slope	SRTM DEM [23]
Aspect	SRTM DEM [23]
Soil type	SoilGrids 2.0 [24]

Sentinel-2 multispectral imagery revisits every 5 days with both satellites, MODIS LST is available daily and as 8-day composites, MODIS NDVI is produced as 16-day composites with phased Terra and Aqua products offset by 8 days, and SMAP soil moisture retrievals occur every 2–3 days with separate AM and PM passes. The temporal alignment step presented in section B.1.a defines how irregular satellite observations were mapped onto the daily modeling framework, but it does not guarantee complete daily coverage. Since the in-situ soil moisture ground truth uses a daily temporal resolution, to address gaps arising from the sparse or irregular satellite observations, we developed a temporal imputation strategy that uses an ensemble of imputers to reconstruct values that were still unavailable after temporal alignment.

The ensemble included ten complementary imputers, each designed to address different missingness patterns in the data. For example, linear interpolation captures short, smooth gaps between nearby observations, while forward and backward filling provide stable estimates over short windows. Rolling averages reduce noise and provide locally smoothed estimates over short windows. Seasonal climatology captures recurring long-term patterns based on historical trends. Regression-based models leverage relationships between correlated features to estimate missing values. Gaussian processes [25] provide flexible, uncertainty-aware interpolation for irregularly sampled signals, and gradient boosted trees [14] capture nonlinear relationships across features for more complex reconstruction tasks. Each imputer i produced an interpolated value at time t , denoted by $x_t^{(i)}$ and an associated confidence $c_t^{(i)}$. The final interpolated value at time t , denoted by x_t is computed as a weighted combination, using the following formula:

$$x_t = \frac{\sum_{i=1}^N w_t^{(i)} x_t^{(i)}}{\sum_{i=1}^N w_t^{(i)}}, w_t^{(i)} = b_i c_t^{(i)} \quad (1)$$

where i is the index for imputer (e.g. $i = 1$ refers to imputer 1 – linear interpolation), $N = 7$ as there are 7 imputers in the ensemble, and b_i is predefined weight for imputer i . The confidence term $c_t^{(i)}$ reflects the reliability of imputer i at time t , and is computed as a function of (i) the temporal distance to the nearest observed value and (ii) imputer-specific reliability heuristics (e.g. gap length for interpolation-based methods). Specifically, predictions closer in time to observed data are assigned higher confidence, while confidence decays with increasing gap size.

Higher weights were assigned to model-based imputers capable of leveraging cross-feature relationships, including linear models (0.25), spline interpolation (0.20), and Gaussian processes (0.15). Moderate weights were assigned to nonlinear and local learning methods such as gradient boosted trees (0.10) and k-nearest neighbors (0.08). Simpler temporal methods, including forward/backward filling (0.07), linear interpolation (0.06), and rolling means (0.05), were assigned lower weights due to their limited ability to capture complex dynamics. Seasonal climatology (0.02) and seasonal naive methods (0.02) received the lowest weights, as they primarily capture long-term trends and may introduce bias in non-stationary conditions. All weights were normalized and the sum of all weights is 1.

The median absolute deviation (MAD), defined as the median of the absolute deviations from the median, was used as a robust measure of variability. At each time step t , we computed the MAD across the predictions of all imputers to identify potential outliers. Predictions that deviate substantially from the median were downweighted in the ensemble, with an outlier factor of 3.0 controlling the sensitivity. This approach reduces the influence of extreme predictions and improves the robustness of the final interpolated value.

Observed values are retained, and only missing entries are imputed. For each feature, a binary mask was generated to indicate whether a value was observed or imputed. Static terrain features were excluded from the ensemble, and missing values were filled by propagating the nearest known values forward and backward. Imputation for dynamic features relies solely on past and neighboring observations, without using future values to avoid information leakage. Consequently, some missing values near the beginning of the time series remain unfilled due to the absence of prior observations.

c) Data Smoothing

Following imputation, a Whittaker smoothing step was applied to selected time-series features such as NDVI, using a regularization parameter of $\lambda = 5000$, to reduce short-term noise while preserving long-term temporal trends. The smoothing procedure leveraged the binary mask columns generated during imputation, assigning full trust to observed values and applying smoothing only to missing or previously imputed entries. This approach preserves the original observations while regularizing reconstructed segments.

2) Features

a) Feature engineering

To capture the temporal dynamics of soil moisture, we performed domain-driven feature engineering by applying structured transformations to raw satellite and weather data listed in Table I. The resulting features were organized into families, where each family groups variables that encode a specific type of temporal behavior or physical process relevant to soil moisture evolution. This grouping improves interpretability by associating features with distinct temporal processes and provides a structured framework for organizing the feature set.

Specifically, the feature families are defined as follows:

- **Change dynamics:** Features that describe short-term variations in soil moisture drivers, including first-order differences, second-order differences, temporal gradients, and percentage changes, capturing rapid wetting and drying processes.
- **Volatility:** Features that quantify variability and local trends over time, such as rolling means, rolling standard deviations, rolling ranges, and exponential moving averages.
- **Memory:** Features that encode persistence effects through lagged observations (e.g. lag-1, lag-7, lag-30, where lag- k denotes the value k time steps before the current observation) and exponentially weighted

memory terms such as the soil moisture memory index, reflecting the temporal autocorrelation of soil moisture.

- **Seasonality:** Features representing periodic or climatological patterns, including sine/cosine encodings of day-of-year, z-score anomalies, seasonal anomalies, and temperature anomaly terms.
- **Meteorological:** Physically meaningful hydrological drivers derived from weather data, including rainfall accumulation over 3-, 7-, and 30-day windows, antecedent precipitation index (API), and days-since-last-rain indicators.
- **SMAP-derived:** Features derived from SMAP mission observations, including combined AM/PM surface soil moisture, AM-PM difference, observability masks, and temporal transformations such as lags, first differences, gradients, percent changes, and rolling summary statistics.

This process resulted in a high-dimensional feature space of 400 features with intentional redundancy across temporal scales. Such redundancy was introduced to capture multi-scale temporal patterns and interactions that may not be represented by a single transformation. However, to prevent overfitting and improve model efficiency, a controlled feature selection step was applied to reduce the feature set prior to model training.

b) Feature selection

Feature selection was conducted using a three-stage pipeline: (1) filter-based screening, (2) sparse regression, and (3) stability-based refinement. First, in the filter-based screening stage, starting from an initial set of 400 candidate features, mutual information was used to retain the top 120 features most strongly associated with the target variable. Second, in the sparse regression stage, Elastic Net was applied to this reduced set to obtain an intermediate subset of 60 features, promoting sparsity while accounting for correlations among predictors. Third, to improve robustness, a stability selection procedure was performed on this 60-feature subset using bootstrap resampling with Elastic Net as the base selector. Specifically, 10 bootstrap samples were drawn using a sampling fraction of 0.8, and features were retained only if selected in at least 60% of runs. The final subset was capped at the top 40 features.

The resulting feature subsets were evaluated using multiple predictive models, including linear regression, XGBoost, and random forest, to ensure robustness across model classes. Performance was assessed using cross-validated coefficient of determination (R^2), along with additional criteria that penalized instability, overfitting, and feature set size. Candidate subsets of up to 40 features (identified from the prior three-stage pipeline) were ranked based on this multi-objective evaluation, and the top-performing subset was selected as a compact and high-quality candidate set. The 40 features remained after this multi-objective evaluation. Starting from this subset, automated ablation studies were performed by iteratively removing individual features and evaluating the impact on model performance, allowing us to quantify each feature's marginal contribution and identify redundant or low-impact features. In

parallel, correlation analysis was used to detect highly collinear predictors, further improving parsimony without sacrificing predictive performance. This final refinement step resulted in a feature set of 38 features spanning six groups.

- Six SMAP-derived soil moisture features (e.g., SMAP_sm_pm_interp_ema02) are satellite-derived soil moisture variables, with suffixes indicating interpolation (interp), and exponential moving averages (emaX) over X observations, etc.
- Thirteen meteorological variables (e.g., G_API, G_rain_sum_3d) are gauge-based measurements, where G_ denotes ground observations, and suffixes such as sum_Xd indicate cumulative values over the past X days.
- Five temporal statistics (e.g., V_rollmin_LST_modis_kobs30) are variability or statistical measures over time, with V_ denoting a transformed statistic, and kobsX indicating the number of past observations used.
- Three lag features (e.g., C_lag_LST_modis_kobs6) represent lagged values, with C_lag_ indicating the variable from X previous observations.
- Four seasonality indicators (e.g., D_sin_DOY, cos_year) encode periodic temporal patterns using sine or cosine transformations of day-of-year or fractional year, with D_ prefix indicating daily resolution.
- Seven static landscape characteristics (e.g., slope, elev) represent unchanging terrain or soil properties, such as slope, elevation, or aspect.

3) Model training

The base model was implemented using a gradient-boosted decision tree (XGBoost) for regression. This approach was chosen for its ability to capture nonlinear relationships, handle heterogeneous feature types, and remain robust to feature scaling and moderate feature redundancy. The training configuration is shown in Table II. The pseudo-Huber objective (i.e. loss function) was selected to improve robustness to outliers and heavy-tailed residuals, which are common in environmental time series.

The dataset was split along two dimensions to support model training. For temporal modeling, the data was partitioned using a strict time-based split to preserve temporal structure and avoid information leakage. The training period spanned January 2017 to December 2020, the validation period from January 2021 to December 2022, and the test period from January 2023 to December 2025. For spatial modeling, a leave-one-station-out (LOSO) approach was applied, in which data from each station was held out in turn while the model was trained on the remaining four stations in WA.

To account for temporal drift and emphasize more recent observations, a weighting scheme was employed for temporal modeling to assign a weight to each training sample based on

its year. The weight of each past year i was computed using the following formula:

$$w_i = \exp(\beta \times (year_i - year_{max})) \quad (2)$$

where $\beta = 0.2$, $year_{max}$ corresponds to the most recent year in the training data, and $year_i$ refers to a past year. This weighting mechanism ensures that the most recent year gets a weight of 1 (i.e. $\exp(0)$), older years have gradually decreased weight – the older year, the smaller weight. The weights were normalized to have a unit mean, ensuring that the overall scale of the loss function remained unchanged. This formulation increases the influence of recent observations while retaining information from earlier periods. The weights were incorporated through the `sample_weight` parameter in XGBoost.

4) Model evaluation

This study selected six evaluation metrics to quantify the performance of our work on soil moisture estimation. The evaluation metrics used were the coefficient of determination (R^2), mean absolute error (MAE), root mean square error (RMSE), bias, unbiased RMSE (ubRMSE), quantile-based error measure. The calculation formulas are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (4)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (5)$$

$$Bias = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \quad (6)$$

$$ubRMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n ((\hat{y}_i - y_i) - Bias)^2} \quad (7)$$

$$\text{quantile-based error measure} = Q_q(\hat{y}_i - y_i) \quad (8)$$

where y_i is the actual soil moisture value, $\hat{y}_i$ is the predicted

soil moisture value, $\bar{y}$ is the mean of the actual soil moisture values, n is the number of observations, and Q_q represents the q -th quantile of the residuals.

III. LORA-BASED SENSOR NODE

A. In-Situ Sensor Node Design

LoRa (Long Range) is a low-power, long-range wireless communication technology that is well suited for resource-constrained environments, where energy efficiency, low cost, and wide area coverage are important design requirements. Its ability to transmit small amounts of data over long distances while consuming relatively little power makes it especially useful for remote environmental monitoring applications.

The sensor node as shown in Figure 1 comprises a custom power-supervisor printed circuit board (PCB), a Heltec WiFi LoRa ESP32 (V3) board, a capacitive soil moisture sensor, an external 16-bit analog-to-digital converter (ADC), a 1 W solar panel, an onboard Lithium Polymer (Li-Po) battery, and a weatherproof sensor housing.

The custom-designed PCB acts as a dedicated power supervisor to support autonomous, in-situ deployment and reduce the impact of the ESP32's relatively high active-state power consumption. Solar input is managed by a dedicated Li-Po charging integrated circuit (IC) that regulates current to safely charge the onboard battery. The custom PCB provides efficient and stable voltage regulation through a combination of switching regulators and low-dropout regulators, while an ATTINY microcontroller gates power to the Heltec WiFi LoRa 32 (V3) board during sleep cycles to reduce parasitic drain to microampere levels. The Heltec board, which integrates an ESP32S3 and SX1262 LoRa module, serves as the primary computation and communication node and focuses on sensor data acquisition and LoRaWAN Medium Access Control (MAC)-layer operations.

The system is enclosed in a weatherproof housing sealed with epoxy resin to protect against moisture, oxidation, and temperature fluctuations during field deployment. For remote-autonomous operation, the sensor node is powered by a 1 W monocrystalline solar panel and onboard Li-Po battery. Soil moisture is measured using a capacitive sensor, while an external ADS1115 16-bit ADC provides higher resolution and more stable readings than the ESP32's internal ADC, improving measurement precision for calibration and monitoring.

The proposed sensor node is designed as a low-cost alternative with an estimated cost of about \$40 USD for a

TABLE II PARAMETER SETTINGS IN TRAINING XGBOOST

Parameter	Value
Objective	reg: pseudohubererror
Number of estimators	5500
Learning rate	0.04
Maximum depth	8
Subsample ratio	0.9
Column sampling	0.8
Minimum child weight	2
L2 regularization (λ)	1.5
L1 regularization (α)	0.03

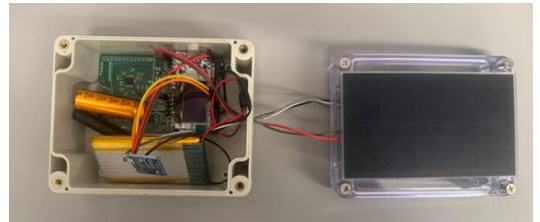


Fig. 1. LoRa-enabled Soil Moisture Sensor Node in a Weatherproof Housing

battery-powered version and about \$70 USD for a solar-assisted version with supervisory power management. While simple hobby-grade capacitive probes may cost only a few dollars, they typically do not provide a complete autonomous field node with wireless communication and power management. In contrast, commercial wireless and research-grade soil-moisture systems commonly range from roughly \$139 to more than \$270 per unit, with some integrated wireless stations costing several hundred dollars or more. This makes the proposed design attractive for resource-constrained and large-area deployments where scalability and spatial coverage are key considerations.

B. Communication Network

The sensor nodes communicate through the Helium platform, which functions as the LoRaWAN Network Server (LNS) and provides a managed interface for registering and authenticating end devices through Over-the-Air Activation (OTAA). By relying on Helium’s decentralized, community-operated gateway infrastructure, the system avoids the cost and maintenance burden of deploying a private LoRaWAN gateway network. Each uplink incurs a small charge in Data Credits (DC), after which validated packets are forwarded by the Helium console to the project server through a secure HTTP (Hypertext Transfer Protocol) webhook.

The backend is hosted on a Virtual Private Server (VPS) configured with a standard LAMP stack comprising Linux, Apache, MySQL, and PHP (Hypertext PreProcessor), and secured with an SSL certificate to ensure encrypted HTTPS transmission and preserve data integrity. This server acts as the central hub for data storage and visualization, with the MySQL database retaining both raw ADC measurements and derived soil moisture percentages for long-term monitoring and trend analysis. At the node level, the soil sensor reading is acquired as a 16-bit raw ADC value representing the soil’s dielectric response, then packaged into a compact binary payload to reduce airtime and power consumption before transmission over the US915 LoRaWAN band. Once received by the Helium network, the payload is Base64-encoded and embedded in a JSON (JavaScript Object Notation) object that includes metadata such as the device EUI (Extended Unique Identifier), RSSI (Received Signal Strength Indicator, and timestamp, and is subsequently delivered to the server webhook for decoding and storage.

C. Experimental Setup

For the in-situ sensor nodes, the primary objective of the initial deployment phase was to validate the complete end-to-end telemetry pipeline. Sensors were deployed at residential properties in the Greater Seattle area, at different topographical elevations and varying distances from Helium network gateways, to verify signal penetration and hardware durability.

D. Data preprocessing

1) Ground Sensor Data Calibration

After the raw JSON payload is received through the webhook, the server executes a PHP script to decode and process the sensor data in three stages. First, the Base64-encoded payload is converted back into raw bytes, and the first two bytes are combined in big-endian order to reconstruct the original 16-bit analog-to-digital converter (ADC) reading.

Second, the server retrieves a device-specific calibration profile from a text file containing the device identifier together with its corresponding wet and dry reference bounds. These calibration values are used to interpret the incoming reading relative to the expected operating range of that sensor. Finally, the raw reading is converted into a soil moisture percentage using a linear interpolation formula as in (9):

$$P = 100 \times \left(1 - \frac{raw - min}{max - min}\right) \quad (9)$$

where P is the moisture percentage, raw is the 16-bit ADC reading, min is the calibrated wet value obtained from fully saturated, gravity-drained soil, and max is the calibrated dry value obtained from oven-dried soil. To prevent invalid outputs and division-by-zero errors, the result is clamped to the range 0–100%.

IV. RESULTS AND DISCUSSION

A. Temporal performance

The XGBoost model was evaluated on the test dataset (January 2023 – December 2025). We also trained and evaluated two other models on the same dataset, including Decision Tree and Gradient Boosting. Table III summarizes the predictive performance across multiple evaluation metrics, including R^2 , MAE, RMSE, ubRMSE, bias, and the quantile-based error measure (Q_{90}).

As shown in table III, the XGBoost model demonstrates the best overall temporal performance than the other two models. It achieved an R^2 of 0.822, which indicates that it captures a substantial portion of the temporal variability. Error metrics are low, with MAE of 0.0283 and RMSE of 0.0397. The near equality of RMSE and ubRMSE (0.0396), along with the small bias (-0.0032), suggests that prediction errors are largely random, with little systematic over- or underestimation. High-quantile error ($Q_{90} = 0.0615$) indicates that large prediction errors remain relatively moderate in magnitude. Overall, these results indicate that XGBoost provides accurate and reliable temporal predictions. This strong temporal performance is likely due to the persistence of soil moisture dynamics, where gradual drying and lagged responses to precipitation create strong temporal dependencies that the model can exploit.

To assess the impact of the temporal weighting mechanism described in section B.3, the XGBoost model incorporating

TABLE III TEMPORAL PERFORMANCE OF XGBOOST, DECISION TREE AND GRADIENT BOOSTING

Metric	XGBoost	Decision Tree	Gradient Boosting
R^2	0.822	0.6764	0.8056
MAE	0.0283	0.0390	0.0306
RMSE	0.0397	0.0546	0.0423
ubRMSE	0.0396	0.0546	0.0421
Bias	-0.0032	+0.0013	-0.0042
Q_{90}	0.0615	0.0969	0.0663

temporal weighting was compared against a version trained without temporal weighting, using identical training hyperparameters. The results are shown in Table IV, demonstrating that the model with temporal weighting achieved slightly higher R^2 and lower errors compared to the model without temporal weighting.

Utilizing temporal weighting in the model led to modest improvements in the model's temporal performance. R^2 increased from 0.814 to 0.822, while MAE and RMSE decreased from 0.0304 to 0.0283 and from 0.0406 to 0.0397, respectively. High-quantile error (Q_{90}) also decreased from 0.0634 to 0.0615, indicating a slight reduction in the magnitude of large prediction errors. The modest improvement is likely due to the model with temporal weighting can give more emphasis to recent observations, which better reflect recent soil moisture dynamics and help adapt to temporal changes in the system.

B. Spatial performance

We evaluated the XGBoost model's spatial performance, since it achieved better temporal performance than the other models evaluated. The model was evaluated using LOSO on five stations in WA state to evaluate the spatial generalizability, which performance is shown in Table V. Spatial performance of the XGBoost model varied across stations. R^2 ranged from 0.403 at Touchet_WA_824 to 0.730 at Darrington, while MAE ranged from 0.0439 to 0.0586 and RMSE from 0.0529 to 0.0677. The value of ubRMSE was slightly lower than RMSE across all stations, and bias values were generally small, ranging from -0.0322 to +0.0262. This suggests that part of the total error comes from consistent bias, while most errors fluctuate randomly around observed values. Stations with higher R^2 (Darrington, Spokane) had lower errors and more accurate predictions, whereas stations with lower R^2 (Touchet, Quinault, SourdoughGulch) exhibited larger and more variable errors. This implies that while the model captures temporal dynamics effectively, spatial heterogeneity and local conditions reduce predictive skill at some stations.

While the model was trained using stations in WA, we further evaluated the spatial performance at selected stations in neighboring states to explore the model's spatial generalizability in completely new environments with different climate and soil characteristics. Table VI summarizes the spatial performance evaluated on locations spanning several western U.S. states, including Oregon (OR), Idaho (ID), Montana (MT), California (CA), Wyoming (WY), and Colorado (CO). The performance was highest at stations like OR_Riley_10_WSW and ID_Murphy_10_W with conditions similar to WA and lower at stations like WY_Lander_11_SSE

TABLE IV PERFORMANCE COMPARISON BETWEEN XGBOOST MODEL WITHOUT TEMPORAL WEIGHTING AND WITH TEMPORAL WEIGHTING

Metric	Without temporal weighting	With temporal weighting
R^2	0.814	0.822
MAE	0.0304	0.0283
RMSE	0.0406	0.0397
ubRMSE	0.0405	0.0396
Bias	-0.0028	-0.0032
Q_{90}	0.0634	0.0615

and MT_Wolf_Point_29_ENE with differing climate or terrain from WA, highlighting both the robustness and the limitations of the model outside the primary study area.

C. Feature importance

To understand the complex relationship between input features and the soil moisture values, as well as identifying the most influential features, we computed SHAP values for each feature for feature importance analysis to quantify the contribution of individual feature to soil moisture estimation. Figure 2 shows the distribution of the SHAP values for different features in the model.

The top 2 features V_rollmin_LST_modis_kobs30 and SMAP_sm_pm_interp_ema02 are the most influential features in the model, which indicates that the model's predictions are primarily driven by land surface temperatures and soil moisture from SMAP. Seasonal signals (e.g. D_sin_DOY, year_frac) and antecedent precipitation metrics (e.g. API_x_year, G_rain_sum_3d) further highlight the importance of temporal cycles and rainfall memory in soil moisture estimation.

TABLE V SPATIAL PERFORMANCE OF THE MODEL XGBOOST EVALUATED USING LOSO ON STATIONS IN WA

Station	R^2	MAE	RMSE	Ub RMSE	Bias	Q_{90}
Darrington	0.730	0.0446	0.0543	0.0539	+ 0.0064	0.0850
Quinault	0.498	0.0439	0.0529	0.0449	- 0.0279	0.0857
SourdoughGulch_WA_985	0.545	0.0518	0.0646	0.0591	+ 0.0262	0.0973
Spokane-17-SSW	0.728	0.0499	0.0577	0.0570	- 0.0092	0.0853
Touchet_WA_824	0.403	0.0586	0.0677	0.0595	- 0.0322	0.1071

TABLE VI SPATIAL PERFORMANCE OF THE MODEL XGBOOST EVALUATED ON STATIONS OUTSIDE WA

Station	R^2	MAE	RMSE	Ub RMSE	Bias	Q_{90}
OR_Riley_10_WSW	0.614	0.0323	0.0422	0.039	-0.016	0.0621
ID_Murphy_10_W	0.608	0.0356	0.0472	0.0465	-0.0079	0.0734
CA_Redding_12_WNW	0.497	0.0435	0.0543	0.0383	-0.0385	0.0812
CO_Boulder_14_W	0.431	0.0454	0.055	0.0476	-0.0275	0.0855
WY_Lander_11_SSE	0.358	0.0404	0.0482	0.0357	+ 0.0324	0.0798
MT_Wolf_Point_29_ENE	0.209	0.0543	0.0683	0.0447	+ 0.0516	0.1044

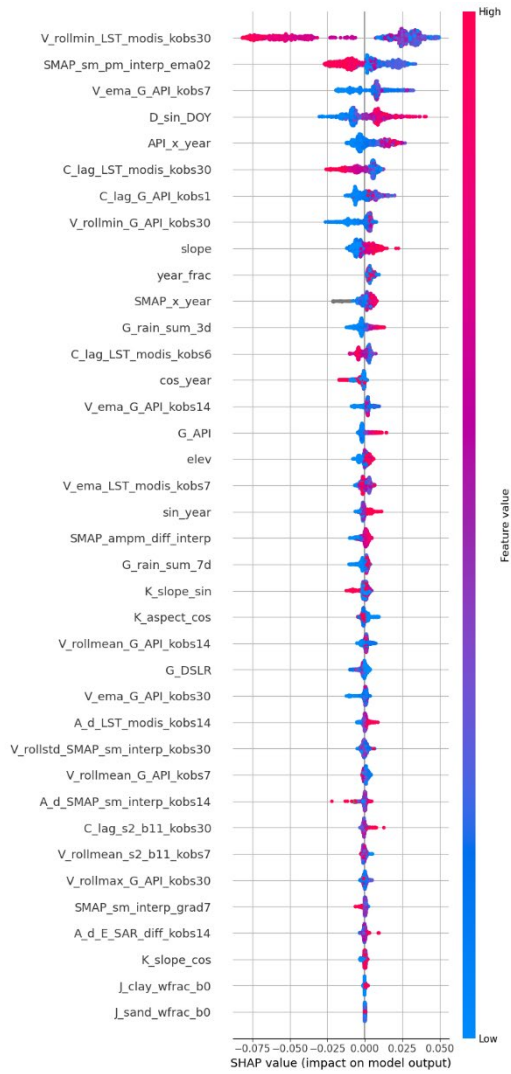


Fig.2. Distribution of feature importance based on SHAP

While observations from SMAP are highly informative, their coarse spatial resolution and limited temporal coverage limit standalone use. By integrating SMAP data with temperature, precipitation history, and terrain information, the model provides more spatially detailed and temporally continuous estimates. Overall, the feature importance is consistent with physical understanding, reflecting temperature, water availability, and seasonal timing which are the main drivers of land surface processes. This indicates that the model is capturing meaningful physical relationships.

D. In-Situ Sensor Observations

The initial in-situ sensor node deployment phase was primarily intended to validate the end-to-end telemetry pipeline rather than to draw broad conclusions about soil behavior. Figure 3 illustrates soil moisture variations at one of the three locations, reflecting frequent occurrences of rainfall over the 7-day monitoring period. For all three locations, the custom PCB power supervisor and ESP32-based node could reliably join the

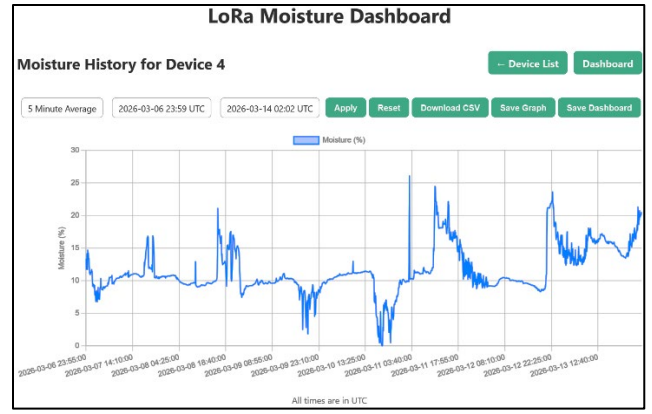


Fig. 3. Temporal Variation of Soil Moisture (%) recorded by a LoRa sensor node (5-Minute Averages over a 7-day Monitoring period)

network through OTAA and sustain uplink communication under real-world weather conditions, thereby confirming successful sensor-to-server telemetry and providing helpful insights into deployment-related factors. A notable site-specific observation was recorded at a location with overgrown grass, where soil moisture values remained consistently higher than at more exposed sites. This behavior is likely attributable to the insulating effect of dense vegetation, which can reduce evaporation and moderate soil temperature, and it served as an indication that the sensor node was sensitive to microclimate variation. The deployment also exposed operational constraints: during periods of extreme low temperatures, including below-freezing conditions, and reduced solar availability, battery life was limited to approximately one week. Even under these challenging conditions, one node maintained stable and consistent reporting. Taken together, the main outcome of this phase was confirmation of reliable end-to-end telemetry in outdoor settings, along with identification of environmental and power-related factors that influence long-term deployment performance.

V. CONCLUSIONS AND FUTURE WORK

In this study, we developed a satellite- and weather-driven machine learning approach to estimate daily soil moisture at locations without ground sensors, integrating diverse satellite observations, meteorological data, and a data imputation strategy to enhance both spatial and temporal coverage. The XGBoost model demonstrated strong temporal performance, achieving $R^2 = 0.822$, $MAE = 0.0283$, $RMSE = 0.0397$, $ubRMSE = 0.0396$, $bias = -0.0032$, and $Q_{90} = 0.0615$. Temporal weighting provided modest improvements by emphasizing recent observations to better capture short-term soil moisture dynamics.

Spatial evaluation across stations in WA showed moderate performance, with R^2 ranging from 0.403 to 0.730 and slightly higher errors than temporal evaluation, reflecting variability in local soil, terrain, and climate conditions. Additional evaluations at out-of-state stations, including Oregon, Idaho, Montana, California, Wyoming, and Colorado, revealed that model performance depends strongly on similarity to the training domain. These findings indicate that while the model effectively captures temporal soil moisture dynamics, its

transferability across heterogeneous regions is limited without more geographically diverse training data.

In this study, we also designed low-cost long-range in-situ soil moisture sensors that communicate over the public Helium network. We successfully demonstrated end-to-end soil moisture telemetry in outdoor residential environments, confirming the feasibility of the hardware, communication, and backend data pipeline. The deployment validated reliable sensor-to-server transmission and provided useful insights into practical field conditions, including microclimate variation, below-freezing temperatures, and limited-sunlight power constraints.

Future work will focus on improving spatial generalizability by incorporating more geographically diverse training data and exploring regime-based modeling approaches to better capture dry, transitional, and wet soil conditions. We also plan to develop time-series models to further enhance temporal prediction and expand field deployments of ground-based sensors to provide additional ground-truth data for model validation in real-world environments. The low-cost sensors will be deployed across diverse terrains in WA state, enabling evaluation across different soil types, microclimates, vegetation cover, and landscape conditions. We plan to use a more robust volumetric water content (VWC) calibration procedure, verifying sensor performance against research-grade TDR/FDR measurements.

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